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Methodology for calculation and verification of the carbon footprint of light means of transport batteries (CFB-LMT)

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Abstract

This draft report provides a first suggestion for the methodological guidelines for calculation and verification of the carbon footprint of light means of transport batteries (CFB-LMT), in line with the requirements as in Article 7 of Regulation (EU) 2023/1542 (Batteries Regulation). It targets exclusively light means of transport (LMT) batteries i.e., batteries that are sealed, weigh 25 kg or less and are specifically designed to provide electric power for the traction of wheeled vehicles that can be powered by an electric motor alone or by a combination of motor and human power, and that are not electric vehicle batteries nor starter lighting ignition (SLI) batteries.

The present draft report is based on the JRC report on rules for the calculation of the Carbon Footprint of Industrial Batteries without external storage (CFB-IND), as published in April 2025[1]. To harmonize the CFB methodologies as far as possible, maximum agreement with the CFB-IND was aspired, and the present document therefore explicitly describes only those parts where a deviation is expected ('delta-document'). All sections deemed identical are deliberately left empty.

The present report has to be considered as a first suggestion for the methodology, to be distributed before the initial stakeholder consultation on the CFB-LMT. It is therefore not a consolidated document, but rather a proposal intended as basis for further discussion and for receiving feedback on the draft methodology.

Specific questions where feedback is sought are scattered throughout the document (green text colour): these are the aspects that will be prioritised for discussion in the corresponding stakeholder workshop.

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[1] European Commission: Joint Research Centre, ANDREASI BASSI, S., ARDENTE, F., CANDELARESI, D., EYNARD, U., FERRONATO, N. and PETERS, J., Rules for the calculation of the Carbon Footprint of Industrial Batteries without external storage (CFB-IND), Publications Office of the European Union, Luxembourg, 2025, https://data.europa.eu/doi/10.2760/6346639, JRC141282.

1 Introduction

The EU Batteries Regulation (Regulation (EU) 2023/1542), published the 28th of July 2023 (European Union, 2023) sets rules on sustainability, performance, safety, collection, recycling, and second life of batteries as well as on information requirements about batteries. Among these, there is the carbon footprint of the batteries (CFB) to be determined and communicated as specified in Article 7 and in the corresponding Annex II of Regulation (EU) 2023/1542. The CFB quantifies the total amount of greenhouse gases emitted along the life-cycle of the battery, in terms of kg of CO_2 equivalent per functional unit. The functional unit is defined by the Annex II of the Regulation as one kWh (kilowatt-hour) of the total energy provided by the battery system over the battery's service life, measured in kWh.

According to Article 7, the CFB shall be declared for "*rechargeable industrial batteries with a capacity above 2 kWh, light means of transport (LMT) batteries and electric vehicle (EV) batteries placed on the Union market*". CFB declarations will then be used to define the CFB performance classes and CFB thresholds that batteries entering the European market shall comply with. Ultimately, Regulation (EU) 2023/1542 will ensure that the expected massive deployment of batteries will be associated with minimum overall carbon emissions. The CFB shall be accompanied by a public version and a non-public version of the CFB supporting study.

The Joint Research Centre (JRC) of the European Commission is providing technical support to the development of the secondary legislation on the CFB, in line with the requirements of the Article 7 and Annex II of Regulation (EU) 2023/1542. The present document provides a first suggestion for the methodology and the technical and scientific base for determining the carbon footprint of light means of transport batteries (hereafter abbreviated as CFB-LMT). As such, it is based on the JRC report that provides the methodological guidelines on the carbon footprint for industrial batteries without external storage (CFB-IND)[2] . All Sections that are expected to remain identical between CFB-LMT and CFB-IND are kept empty, focusing on the aspects that differ between IND and LMT batteries. Some sections deviate only partially: in these the text passages that are identical to the corresponding part in the CFB-IND Report are written in **blue font colour**.

The present report is a working document to be used as basis for the discussion with stakeholders and policymakers. Specific questions on which feedback is explicitly sought are scattered throughout the document, highlighted by **green font colour**. However, stakeholders are explicitly encouraged to comment on any aspect of the document not limiting the feedback to these questions.

Note: The text of carbon footprint rules has been kept as concise as possible. All sections specified as "Note" have to be considered as explicative to the text and not be intended as part of the rules for Carbon Footprint of Batteries (CFB).

Unless otherwise specified in this document, the EF method as in the EC Recommendation 2021/2279 (European Commission, 2021) shall be considered as general guidelines about how to determine the carbon footprint.

[2] https://op.europa.eu/en/publication-detail/-/publication/53ee8e6a-18dc-11f0-b1a3-01aa75ed71a1/language-en

1.1 Terminology

This report uses the following terminology to indicate the requirements, as:

— The term “shall” is used to indicate what is required in order to calculate the CFB.

— The term “may” is used to indicate an option that is permissible. Whenever options are available, the CFB supporting study shall include adequate argumentation and evidence to justify the chosen option.

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2 Scope

This document provides the draft methodological guidelines and the technical and scientific base for calculating the carbon footprint of light means of transport (LMT) batteries within the scope of the Article 7 of Regulation (EU) 2023/1542 i.e., any *"battery that is sealed, weighs 25 kg or less and is specifically designed to provide electric power for the traction of wheeled vehicles that can be powered by an electric motor alone or by a combination of motor and human power"*. The final methodology to be followed under Regulation (EU) 2023/1542 for determining the CFB-LMT will be ultimately defined in the corresponding Delegated Act. The CFB-LMT shall be declared for any LMT battery that falls within the scope of Article 7, independently of its cell chemistry and of the specific application. The scope of the CFB-LMT excludes batteries that have been subject to preparation for re-use, preparation for repurposing, repurposing, or remanufacturing, and that had already been placed on the market or put into service before undergoing such operations (according to the Article 7 (point 5) of Regulation (EU) 2023/1542).

Note: Although the rules defined in this document are generic and applicable to any rechargeable LMT battery independently of its chemistry, the following battery chemistries have been analysed for the testing and development of the present CFB-LMT rules (especially related to Sections 5.2 and 6.3), being currently considered the most relevant ones:

- *NMC-type lithium-ion battery (LIB) chemistries*
- *LFP-type lithium-ion battery (LIB) chemistries*

Other emerging chemistries like Sodium-Ion, solid-state LIB, etc. could potentially become relevant in future, especially once achieving market maturity. The same applies to emerging anode / cathode materials for any of the listed battery chemistries and to other battery technologies. If deemed necessary, the CFB rules may be updated correspondingly.

Question to stakeholders:

- Is there any other battery chemistry of particular relevance that should be considered or analysed more in detail?

LMT batteries may come in two forms:

- ***detachable* batteries**, typically in the range of between 300 and 1000 Wh, mostly 500-600 Wh, 36-48 V, and containing an array of cylindric cells. These are often generic (i.e., not designed for a specific vehicle) and can be mounted in a variety of LMT vehicles, mostly e-bicycles / pedelecs, and e-scooters. They are partially sold individually (i.e., not together with the vehicle). The charger is external and not always sold together with the battery (e.g., not for a replacement battery).

- ***integrated* batteries**, mostly in small electric motorcycles / mopeds or >2 wheelers. With a maximum mass of 25 kg to classify as LMT battery, these have an energy capacity of roughly 3-5 kWh (for a 25 kg battery of 120-200 Wh/kg). While they may be swappable, they are designed for a specific vehicle. The charger might be integrated in the battery or be external to it.

Note: The ongoing discussion on standardisation activities on LMT minimum durability requirements suggest distinguishing between: (i) LMT batteries used in vehicles with human assisted propulsion (such as e-scooters or pedelecs) and (ii) those used in vehicles that are exclusively powered electrically. The rationale is that for the first category the discharge depth is typically higher (until

100% DoD) due to the absence of range anxiety. For the CFB-LMT this approach was not adopted, considering that such aspects are already implicitly considered in the declared lifetime under Article 10 and/or the given warranty.

Question to stakeholders:

- Would you agree that the suggested classification in "detachable" and "integrated" LMT batteries is necessary/brings added value? If yes, explain why (e.g., significant differences in battery layout, battery handling, or end-of-life).
- If you disagree with the classification, which alternative classification would you suggest? Or should rather all LMT batteries be treated equally?
- Would you agree with not distinguishing between LMT batteries used in vehicles with human assisted propulsion (such as e-scooters or pedelecs) and those used in vehicles that are exclusively powered electrically? If not, what are the aspects of distinction that could have a significant impact on the carbon footprint and why?

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3 Calculation Rules

3.1 Life Cycle Impact assessment and Reference Flow

The carbon footprint shall be calculated as the amount of CO_2-equivalents emitted during the life cycle stages of the battery that are within the system boundary, expressed in kilogram ('kg') CO_2-equivalent, divided by the total amount of energy provided by the battery over the battery's service life. It shall be reported in kg CO_2-equivalents/ kilowatt-hour ('kWh'), with a resolution of two significant digits.

The amount of kg CO_2-equivalents emitted shall be determined by identifying the emissions of greenhouse gases related to the materials and energy used and, where relevant, produced in the life cycle stages of the battery that are within the system boundary, in accordance with Sections 4.1 and 4.2. The amount of kg CO_2-equivalents shall be calculated from these emissions of greenhouse gases by applying the Environmental Footprint ('EF') 3.1 impact assessment method available in the Life Cycle Data Network (LCDN) for the impact category "climate change".

All quantitative input and output data collected to quantify the carbon footprint shall be calculated in relation to the reference flow. The reference flow shall be calculated as the total mass of battery divided by the total amount of energy provided by the battery over the battery's service life, and thus measured in kg/kWh.

If, over the course of time, due to changes in the bill of materials (i.e., the list of all material inputs needed to manufacture the product in scope of the study), changes in the origin of the materials, changes in processes, changes related to the use of electricity and other auxiliaries, or any other changes, the CFB increases by more than 10% compared to the carbon footprint calculated, this shall be considered a change to the battery's technical characteristics relevant for the requirements of Regulation (EU) 2023/1542 and thus for the new battery model a new carbon footprint shall be calculated and a new carbon footprint declaration shall be drawn up.

3.2 Functional unit

The **functional unit** is defined (according to Annex II of Regulation (EU) 2023/1542) as "*one kWh (kilowatt-hour) of the **total amount of energy** delivered by the battery over its service life, measured in kWh".*

There are essentially **two options** for determining the total amount of energy, indicated here below in bullet points (a) and (b):

(a) Based on warranty.

The **total amount of energy** provided by the battery over the battery's service life ('E_{total}'), expressed in kWh, shall be calculated as follows:

$$E_{total} = energy\ capacity \cdot FEqC\ per\ year \cdot years\ of\ operation$$

Where:

- ***energy capacity*** is the useable energy capacity of the battery in kWh at the beginning of life, namely the energy available to the user when discharging a new fully charged battery until the discharge limit set by the battery management system;

- ***FEqC per year*** is the typical number of full equivalent charge-discharge cycles per year and equals **30 for all LMT batteries**

Note: this option is inspired by the approach taken in the draft delegated act defining the methodology for determining the CFB of electric vehicle batteries, as published on "Have your Say" for stakeholder consultation in April 2024.

Note: 30 FEqC/y is equal to the value established for Type L electric vehicle batteries (motorcycles) in the draft Delegated Act defining the methodology for calculating the carbon footprint of EV batteries. Using the same value avoids a discontinuity for small motorcycles with batteries around 25 kg that would be considered EV batteries or LMT batteries according to the corresponding battery mass.

Note: Literature review obtained average yearly mileages for e-bikes and pedelecs of between 1600 and 2400 km (see Table 1*). 2000 km/y equal, with a driving range of 65 km per battery charge, 30 FEqC/y. However, for electric mopeds and scooters, ranges are often lower.*

- ***years of operation*** is determined by the commercial warranty according to the following rules:
    - (i) the duration of the warranty on the battery in years applies;
    - (ii) if there is no specific warranty on the battery, but a warranty on a vehicle in which the battery will be used, or parts of a vehicle that include the battery, the duration of that warranty in years applies;
    - (iii) by way of derogation of points i) and ii), if the duration of the warranty is expressed in both years and kilometres whichever one is reached first, the lowest number of the two in years applies. For this purpose, a **conversion factor of 2000 km** equalling one year shall be applied;
    - (iv) if the battery is used in multiple vehicles and the results of the approach in point ii) and, where applicable, iii) would be different between those vehicles, the lowest resulting number in years applies;
    - (v) only warranties that are related to a remaining energy capacity of at least **60% of the useable energy capacity** of the battery in kWh at the beginning of life shall be taken into account in points i) to iv). Warranties that explicitly exclude any individual components that are essential for the proper functioning of the battery or that restrict the use or storage of the battery apart from conditions that are within the typical use of such batteries shall not be taken into account in points i) to iv);
    - (vi) if there is no warranty or only a warranty not compliant with the requirements under point (v), a figure of **six months** shall be used, except for cases where a warranty is not applicable, such as where there is no transfer of ownership of the battery or vehicle, in which case the manufacturer of the battery shall determine the number of years of operation and justify it in the public version of the carbon footprint study.

Note: Detachable LMT batteries typically have warranties of between 6 months and 2 years, and for 500-1000 cycles. The warranty mostly guarantees a remaining energy capacity at EoL of 60%, in some cases 70%.

Note: CFB-EV uses 70% of the initial useable energy capacity as threshold for Type-L vehicles (motorcycles).

Table 1. Average mileages for LMT vehicles obtained from literature

km/year	LMT annual mileage (km)	Type	Source
1600	Annual average mileage (e-bike): 1600 km	e-bike/ pedelec	Bikelix e-bike stats (page recently down)
1949 (med) 2304 (avg)	the median annual self-reported e-bike mileage was 1218 miles, and the mean was 1440 miles.	e-bike/ pedelec	Electric Bikes: Survey and Energy Efficiency Analysis Efficiency Vermont
2400	A service life of eight years and a yearly distance of 2400 km were estimated for an average commuter bike by European Cyclists' Federation (ECF, 2011)	e-bike/ pedelec	Huang et al. (2022) Comparative life cycle assessment of electric bikes for commuting in the UK.
2400	most electric scooters average around 1,000 to 2,000 miles per year.	e-scooter	https://www.levyelectric.com/resources/what-is-high-mileage-for-an-electric-scooter%3F
3208	2005 miles on average for motorcycles (across all sizes)	Motorbikes (all sized)	https://afdc.energy.gov/data/10309
4019	Average annual mileage of sharing e-bikes in Barcelona	e-bike/ pedelec (sharing)	https://www.mdpi.com/2071-1050/14/7/4139

Source: JRC own elaboration.

Question to stakeholders:

- Would you agree on the value of 30 full equivalent charge-discharge cycles (FEqC) per year? If not, what alternative would you suggest? (please note that FEqC would be lower than the number of full cycles or partial cycles, as defined below)

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- Would you agree on the conversion factor of 2000 km per year to account for calendar life? If not, what alternative would you suggest?
- Would you agree on the threshold of 60% remaining energy capacity (with respect to initial energy capacity) as minimum requirement for the warranty?
- Would you agree on the value of 6 months lifetime in case no warranty compliant with the requirements is given?
- Do you have any other concerns regarding the warranty approach? Please specify.
- Apart from these questions, we particularly welcome also primary data on lifetimes, average mileages, warranties, number of cycles and how each of these are determined / declared.

(b) Based on Article 10 declared values.

The **total amount of energy** provided by the battery over the battery's service life ('E_{total}'), expressed in kWh, shall be calculated as follows:

$$E_{total} = initial\ useable\ energy\ capacity \cdot cycles\ provided$$

Where:

— ***Initial useable energy capacity*** is the energy available to the user when discharging a fully charged, new battery (at beginning of its service life) within the charge/discharge limits set by the battery management system under standard operating conditions. "Fully charged" refers to charging until the charge level set by the battery management system under standard operating conditions.

— ***Cycles provided*** is the battery expected lifetime expressed in full equivalent cycles (FEqC), as declared according to Article 10 / Annex IV. The values shall correspond with those reported in the technical specification delivered with the battery.

If the battery expected lifetime is not expressed in FEqC, but in **full cycles** within the charge/discharge limits set by the battery management system under standard operating conditions, the number of FEqC may be estimated by multiplying the number of full cycles with the mean value (arithmetic average) between the initial state of health and the state of health at the end of the battery's service life (linear interpolation).

If the battery expected lifetime is expressed only in **partial cycles**, the number of FEqC may be estimated by multiplying the number of partial cycles with the ratio between the average energy delivered per partial cycle over service life and the initial useable energy capacity.

The exact test procedure (including charge-discharge profiles over time, C-rates, temperature), the standard operating conditions and, if applicable, the underlying standard used for determining the battery expected lifetime shall be provided in the public version of the CFB supporting study. The threshold for determining the end of service life of the battery shall be equal or above 60% of the battery's initial useable energy capacity. "End of service life" refers to the end of the battery's first service life, i.e. within the primary application for which it was originally designed; any second life is not considered.

If the value of '*cycles provided*' is higher than the lifetime in years (as declared according to Article 10 / Annex IV) multiplied with **120 cycles/year**, then a figure equal to 'lifetime in years * 120 cycles/y' shall be used for '*cycles provided*' in the calculation of the total amount of energy.

Note: This impedes too extreme discrepancies between service life in cycles and service life in years. Considering an average of 30 FEqC/y as suggested under Option (a), this is still 4x higher than the typical use frequency.

In case the above-mentioned requirements are not met, a **default value of 250 FEqC** shall apply for 'cycles provided'.

The total amount of energy, the initial useable energy capacity, the lifetime in cycles and in years, and the threshold for determining the end of service life shall be provided in the public version of the CFB supporting study.

Question to stakeholders:

- Would you agree on the consideration of calendar life applying a fixed value for cycles/year? If not, please explain.
- Would you agree with the value of 120 cycles/year in the conditions explained above? If not, what would you suggest?
- Would you agree on the threshold of 60% remaining energy capacity (with respect to initial energy capacity) as minimum requirement for the warranty?
- Would you agree with the use of a default value of 250 FEqC for 'cycles provided' in case the requirements are not met? If not, which alternatives would you suggest?

*Note: **FEqC** is the number of times the battery would theoretically be able to discharge its initial useable energy capacity. The FEqC multiplied with the initial useable energy capacity gives the total energy provided by the battery over lifetime.*

*Note: **Full cycles** is the number of times the battery can fully discharge within the charge/discharge limits set by the battery management system under standard operating conditions. The actual energy delivered in each full cycle decreases over lifetime. For a new battery (at beginning of its service life), one full cycle is equivalent to one FEqC. The full cycles multiplied with the initial useable energy capacity and the average value between 100% and the state of health (or remaining useable energy capacity in %) at EoL gives the total energy provided by the battery over lifetime.*

*Note: **Partial cycles** is the number of times the battery can discharge with a given discharge profile, but without fully discharging within the charge/discharge limits set by the battery management system.*

Note: The boundaries between LMT batteries and Type-L EV batteries are only determined by the battery mass; a battery with a mass of 25.1 kg is an EV battery, while with 25.0 kg it is LMT battery, even if mounted in the same vehicle and providing the same purpose. It is therefore considered desirable to keep the FU as similar ass possible between both, limiting discontinuities.

Question to stakeholders:

- Which of the two suggested approaches would you consider more appropriate and why?
- What is the market relevance of swappable batteries and would any specific provisions be required for these?

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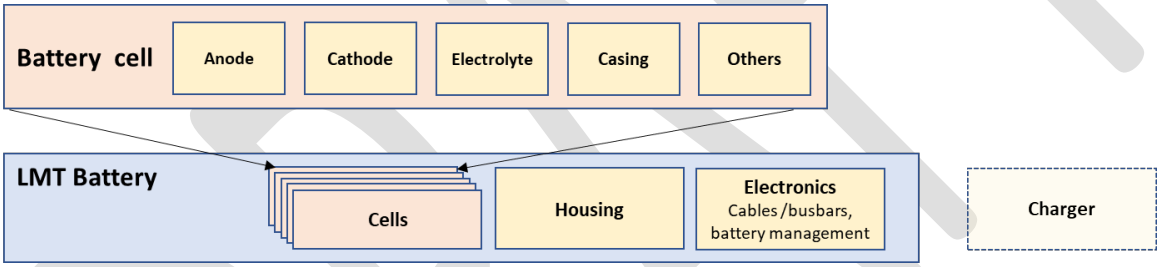
4 System Boundaries and cut-off rules

4.1 System boundaries

The system boundaries define which parts of the product life cycle and which associated life cycle stages and processes belong to the analysed system (i.e., are required for carrying out its function as defined by the functional unit).

Due to their limited size, LMT batteries typically follow a 'cell-to-pack' design, where the battery cells (mostly cylindrical cells) are assembled directly into the final battery. The charger, though often delivered together with the battery, is not contained within the system boundaries. A schematic representation of the considered battery system components is provided in Figure 1. The definition of the system boundaries follows a general supply-chain logic, including all stages from raw material acquisition and pre-processing, production of the main product, product distribution, etc.

Figure 1. Battery system components. The charger is represented with dotted lines because it is outside the system boundaries



Source: JRC own elaboration.

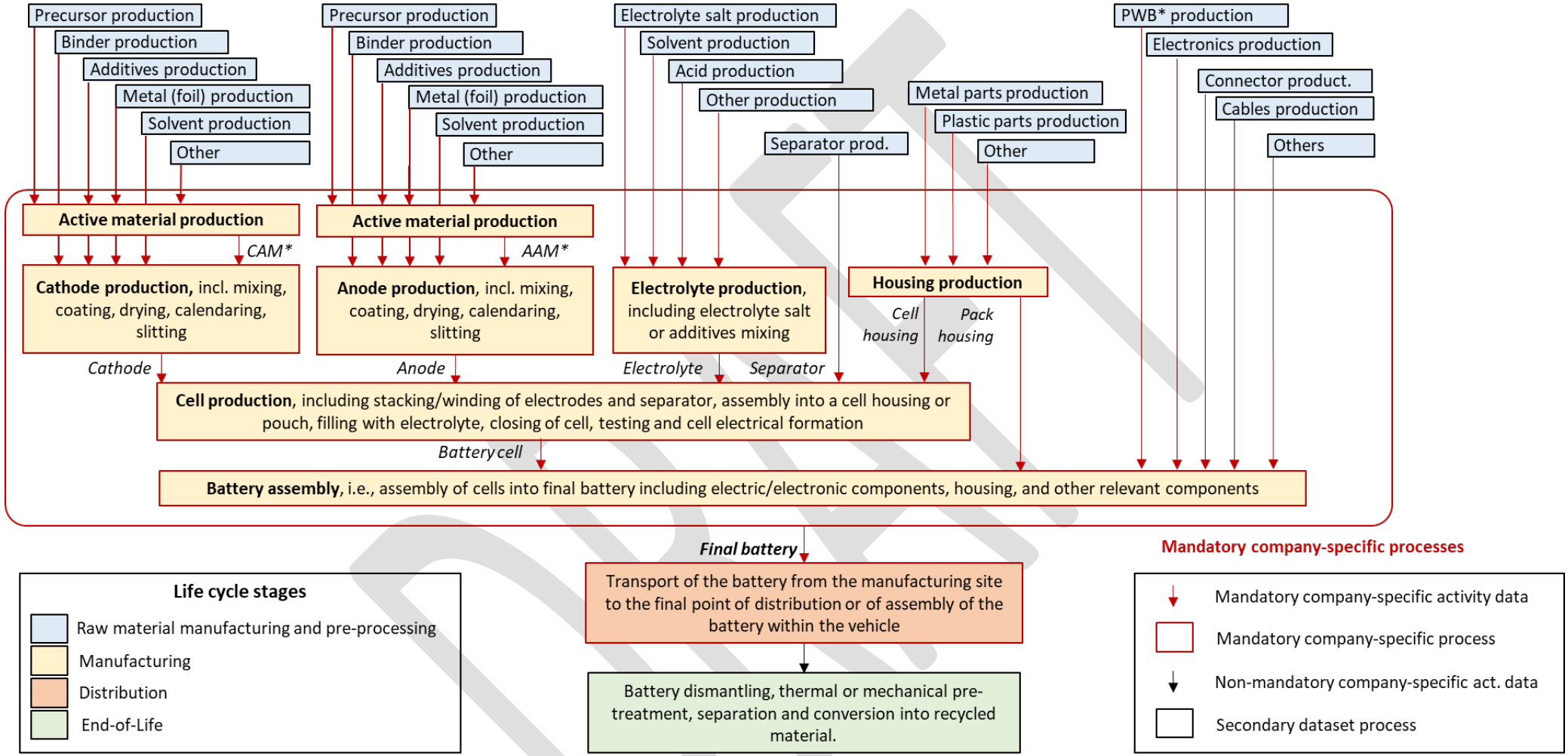
Question to stakeholders:

- Would you agree on excluding the charger from the system boundaries? Might this introduce any (dis-) advantage to certain LMT batteries?
- Would you agree on the system components as depicted in Figure 1? If not, what should be amended?

4.2 Life Cycle Stages

The life cycle stages that shall be included in the system boundary are depicted in Figure 2 and described in the following. Other types of batteries (e.g., anode-free batteries) may have a different visualization of their system boundaries. The squares in Figure 2 represent processes, while arrows represent inputs (activity data such as kg of solvent, kg of additive etc.). The different colours (blue, yellow, orange, and green) indicate to which life cycle stage each process belongs, while red arrows and red borders indicate that a process/activity data shall be company-specific (Section 5). Black arrows and black borders indicate non-mandatory company-specific activity data and processes, respectively. The use-phase of the battery is explicitly excluded from the system boundaries according to Article 7 and Annex II of Regulation (EU) 2023/1542.

Figure 2. System boundaries for a generic LMT battery. Processes and materials may vary depending on the battery type and chemistry



*Source: JRC own elaboration. *PWB: populated Printed Wiring Board. CAM: cathode active material. AAM: anode active material.*

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4.2.1 Raw material acquisition and pre-processing

Identical to CFB-IND.

4.2.2 Main product production

The life cycle stage of the 'Main product production' covers the following activities:

- cathode active material production;
- anode active material production, including the production of graphite and hard carbon from their precursors;
- anode and cathode production, including the mixing of ink components, coating of ink on collectors, drying, calendaring, slitting;
- electrolyte production, such as mixing of electrolyte salts and solvents;
- assembling the cell components into a battery cell, including stacking/winding of electrodes and separator, assembling into a cell housing or pouch, injection of electrolyte, closing of cell, testing and electrical formation;
- assembling the cells into a final battery including electric/electronic components, housing, and other relevant components that are necessary for the safe and reliable operation of the battery;
- transport operations of the intermediate products to the site where they are used.

The modelling of the management of the waste generated during this life cycle stage shall be included in calculation of the carbon footprint of this life cycle stage. More information on how to account for manufacturing scraps and recycled content in the 'Main product production' life cycle stage is provided in Section 6.3.2.

4.2.3 Distribution

Identical to CFB-IND.

4.2.4 End of Life and recycling

Identical to CFB-IND.

4.3 Cut-off rules

The following processes may be excluded from the modelling:

(a) manufacturing of capital goods, including equipment;

(b) production of packaging materials;

(c) any component that is not physically contained in or permanently attached to the housing, such as the charger;

(d) auxiliary inputs to manufacturing plants that are not directly related to the battery production process, including heating and lighting of associated office rooms, secondary services, sales processes, administrative and research departments;

Note: the auxiliary inputs to the 'Main product production' life cycle stage were estimated to be negligible: assuming a consumption of a generic office as 293 kWh/m^2 (Pérez-Lombard, Ortiz and Pout, 2008) *and assuming a production area of 267000 m^2 (compatible with a facility of 20 GWh/year of production capacity), the auxiliary energy consumption is estimated to be about 0.2% of the electrical energy required by the production line.*

A general cut-off of 1% in mass may be applied to material inputs per system component as depicted in Figure 1, by neglecting input and output flows that make up less than 1% to the total mass of the system component.

The cut-off may be applied within the modelling of the production of the following system components in the 'Main product production' life cycle stage:

(a) battery cell anode;

(b) battery cell cathode;

(c) battery cell electrolyte;

(d) battery cell housing;

(e) the battery cell components other than the ones listed in points (a) to (d) combined;

(f) battery housing / enclosure;

(g) battery electronics;

(h) battery other components than the ones listed in points (f) and (g) combined.

Question to stakeholders:

- Would you agree on the list of system components or are there any relevant components missing?
- Are there LMT batteries with a dedicated cooling (or heating) system or thermal management system, especially for LMT batteries operating in environments with extreme temperatures? If yes, please indicate the configuration of the cooling/heating/thermal management system and the type of vehicle where the battery is mounted in (e.g. e-bike vs small motorcycle).

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The cut-off may be applied within the modelling of the following processes in the 'Raw material acquisition and pre-processing' life cycle stage:

(a) mining;

(b) beneficiation or ore processing, from ore to concentrate;

(c) primary extraction, either pyrometallurgical or hydrometallurgical;

(d) refining;

(e) finishing.

Grinding media in the raw material acquisition and pre-processing life cycle stage shall be accounted for even if it might fall under the cut-off criteria.

If a cut-off is applied, the mass gap shall be closed on system component level (in the case of 'main product production' life cycle stage), or on process level (in the case of 'raw material acquisition and pre-processing' life cycle stage), by adding the missing mass to the material input flow with the highest specific carbon footprint among all remaining materials used in the system component to which the cut-off was applied. The carbon footprint study shall mention if and where the cut-off of 1% in mass has been applied.

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5 Data collection requirements and quality requirements

Identical to CFB-IND.

5.1 Mandatory company-specific processes

Identical to CFB-IND

5.2 Non-mandatory company-specific processes

Non-mandatory processes are divided into most relevant and non-most relevant processes depending on their relevance on the overall CFB calculation. Guidelines on the modelling of most relevant non-mandatory company-specific processes are provided in Section 5.2.1, and on the modelling of non-most relevant ones in Section 5.2.2.

The most relevant processes are the following:

- production of the following cathode active material precursors: cobalt, nickel, iron, lithium, (all either metallic or as chemical compounds);
- production of the following anode active material precursors: lithium, (either metallic or as chemical compounds), silicon, graphite and hard carbon precursors (such as natural graphite ore, cokes, etc.);
- production of electrolyte salts and precursors: Lithium hexafluorophosphate ($LiPF_6$), Lithium bis(trifluoromethanesulfonyl)imide (LiTFSI), lithium compounds, sodium compounds, nickel compounds, solid electrolytes;
- production of copper, e.g., in the current collectors, busbars and cables;
- production of aluminium, e.g., in the current collectors, busbars / cables and housing;
- production of steel, e.g., in the housing.

Note: the most relevant processes include all that have been identified for LMT batteries. Examples of non-most relevant processes are the production of the solvent used in cathode and anode, of the carbon fibres used in the housing, of the electronics/PWB, the manufacturing of the plastic for the collector or housing, etc.

Note: Non-most relevant processes shall be modelled (i.e., cannot be neglected). The distinction between most relevant and non-most relevant only affects the possible use of primary data for modelling (possible for most relevant, not possible for non-most relevant.). For instance, production of manganese for cathode active material precursors is not included among the most relevant processes (due to lower specific environmental significance), and shall be modelled following the rules valid for non-most relevant processes.

5.2.1 Modelling requirements of the most relevant processes

Identical to CFB-IND.

5.2.2 Modelling requirements for the non-most relevant (other) processes

Identical to CFB-IND.

5.3 CFB-compliant company-specific datasets

Identical to CFB-IND.

5.4 Requirements for the collection of company-specific Life Cycle Inventory (LCI) data

Identical to CFB-IND.

5.5 Data quality rating (DQR) of the CFB declaration and of CFB-compliant datasets

Identical to CFB-IND.

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6 Inventory data – Modelling requirements

6.1 Electricity modelling

It is recommended that the electricity modelling shall finally align with the requirements laid out in the final DA-EV defining the methodology for calculating the CFB-EV (still under finalisation at the date of the publication of the present report).

6.2 Allocation rules

6.2.1 Allocation of multi-functional processes

Identical to CFB-IND.

6.2.2 Allocation of energy and auxiliary inputs of production lines

Identical to CFB-IND.

6.3 Recycled content and End of Life modelling

The recycled content and the waste generated during all the life cycle stages shall be modelled with the use of the circular footprint formula ('CFF') and shall be reported at the life cycle stage where the waste generation occurs. This section describes how to apply the CFF to the recycled content and to the End of Life and recycling life cycle stage of the battery in scope. For any other case, Section 4.4.8 of Annex I to Commission Recommendation (EU) 2021/2279 shall apply (European Commission, 2021).

The CFF shall not be used for any materials or objects rejected during the battery manufacturing process that are re-used as an integral part in the same process and that do not need to be recycled, such as run-around scrap. However, emissions and process inputs associated with their processing shall be accounted for in the corresponding life cycle stage.

The End of Life and recycling life cycle stage of the battery shall be modelled either:

- with the default battery recycling process specified in this section, using the default CFF parameters provided in Table 2 and the default battery cell recycling process provided in Table 3 and Table 4;
- with a company-specific battery cell recycling process, which may only be modelled for the share of the batteries for which evidence is provided that they will be recycled in a specific recycling plant operating at commercial scale and that the specific recycling process corresponds to the battery model for which the carbon footprint is declared. Such specific recycling process shall be modelled with company specific data in accordance with Section 5.1. Table 2 specifies which default values shall be replaced with company-specific values in such case. In case a company-specific battery cell recycling process is modelled, the deviations from the default battery recycling process and the corresponding evidence shall

be detailed in the carbon footprint study. The justification for applying a company-specific battery cell recycling process shall be provided in the public version of the carbon footprint study.

6.3.1 The circular footprint formula ('CFF') applied to the battery in scope

The Circular Footprint Formula (CFF) describes the allocation of impacts and credits from the EoL handling of the battery. The following rules apply:

- The default recycled content of the materials in the battery in scope shall be zero unless evidence of the traceability throughout the supply-chain is provided in the CFB supporting study.
- The modelling requirements for the 'End of Life' life cycle stage depend on the battery chemistry of the battery in scope (i.e., the battery for which the CFB is developed and declared).
- It is assumed that a fraction of the batteries is not properly returned and treated under EU legislation[3]. By default (and unless declared differently), the properly returned batteries are assumed to be sent to (manual) disassembly / dismantling (Figure 3).
- The outputs of the dismantling are the battery cells and remaining battery components that are sent to individual recycling processes (steel and aluminium from the housing, copper, electronics/PWB for recovering gold, silver, copper and palladium, cables for recovering copper). The sorted plastic is sent to incineration with energy recovery and the rest to landfill. The dismantled battery cells are then sent to the default battery cell recycling process, which varies according to the battery cell chemistry:

  1. The default battery cell recycling process shall be modelled according to Table 3 and Table 4, assuming a combination of a pyrometallurgical and hydrometallurgical treatment (that recovers copper sulphate, nickel sulphate, and cobalt sulphate).
  2. For other battery chemistries that do not fall under any of the previous categories, no established battery cell recycling processes have been identified and therefore the default EoL scenario assumes no battery cell recycling (i.e. no material recovery from battery cells).

  The remaining fraction of the batteries is assumed to be not properly returned and treated under EU legislation (named as "non-properly returned" batteries in Figure 3) and assumed to be roughly dismantled (with a process similar to that for properly returned batteries above, but without carrying out the recycling of the electronics/PWB), whereas battery cells are assumed to be landfilled.

- A company-specific return rate may be used only for the share of batteries covered by an ownership business models where the property of the battery stays with the manufacturer and where contractual evidence of higher return rates is provided, including evidence that the batteries covered under that rate are recycled in accordance with Regulation (EU) 2023/1542.

[3] Article 61 of Regulation (EU) 2023/1542 requires battery producers to "*ensure that all waste batteries [..] are collected separately*". However, this is only possible for batteries that become waste batteries within the EU, and not necessarily for all that have been made available on the market. The return rate considers that there is a small fraction of batteries of unknown whereabout (e.g., exported) that are not captured by Article 61 and therefore not recycled properly under EU Legislation.

— A company-specific battery recycling process may be used only for the share of the batteries for which the CFB declarant is able to provide evidence that the same batteries will be recycled in a specific recycling plant and that the specific recycling process corresponds to the battery model for which the CFB is declared. The corresponding evidence shall be provided in the CFB supporting study.

— The environmental impacts of collecting and dismantling the waste batteries shall not be included in the model (i.e., they are defined as cut-off in Section 4.2.4). The return rate is expressed with the parameters R_{Return} in the CFF, while the efficiency of the dismantling is assumed to be 100%.

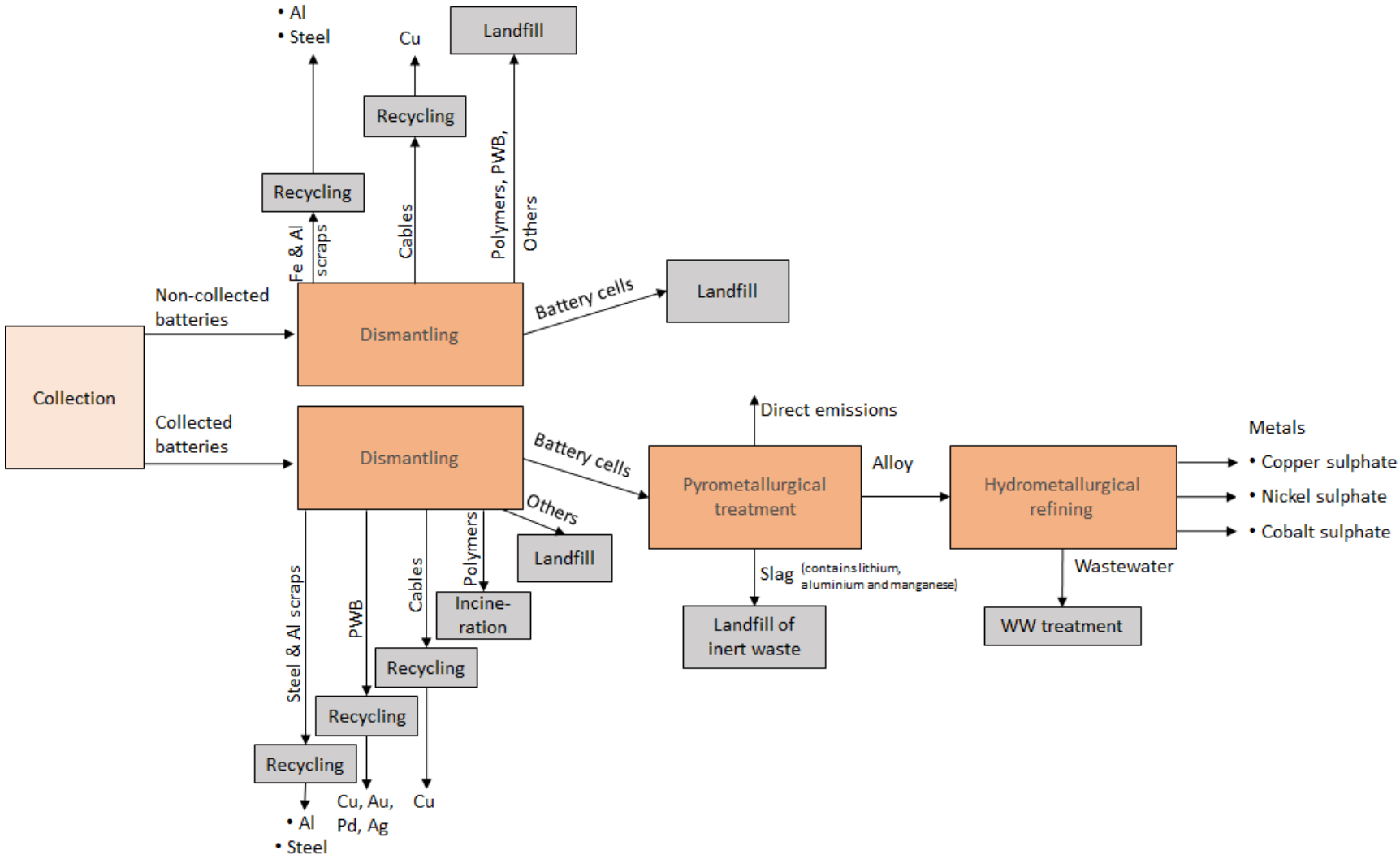
— The incineration with energy recovery only applies to the plastic waste from the dismantling of the properly returned batteries. The energy recovery from plastics in the battery cells is instead accounted for by the default battery cell recycling process.

Question to stakeholders:

- Would you agree on the recycling flowsheet as depicted in Figure 3? If not, which amendments would you suggest?
- Should there be a distinction between detachable and integrated batteries in terms of recycling, specifically regarding the disassembly until cell level? If yes, how?

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Figure 3. Schematic flowsheet of the default 'End of Life' life cycle stage of the battery in scope



Source: JRC own elaboration. PWB: populated Printed Wiring Board.

Note: The flowsheet in Figure 3 depicts the default recycling process, which reflects current industry practice. Other recycling processes may recover other materials (such as lithium). Such processes can be claimed and modelled if corresponding evidence is provided, including the corresponding credits for recovered materials.

The below equations describe how to apply the CFF to the battery in scope. The CFF shall be applied per material, unless specified differently in the description of the single parameters. In particular:

- the impacts of the battery cell recycling shall not be allocated per material but shall be allocated to the full battery using an average allocation factor ($A_{Batterycell}$) per battery cell;
- the impacts of recycling of the electronics/PWB shall not be allocated per material but shall be allocated to the full PWB using an average allocation factor (A_{PWB}) per the PWB.

The CFF is the sum of several terms, each related with different processes associated with the materials that constitute the battery in scope along its life cycle:

- Impacts of using primary and secondary materials ('material input'):
$$\sum_{Mat}[(1-R_{1_Mat})\cdot E_{V_Mat}+R_{1_Mat}\cdot(A_{Mat}\cdot E_{recycled_Mat}+(1-A_{Mat})\cdot E_{V_Mat}\cdot\frac{Q_{Sin_Mat}}{Q_{P_Mat}})]$$
- Impacts and credits of producing secondary materials from dismantling, such as steel and aluminium parts and copper from the cables ('dismantling'):
    - Materials from the properly returned fraction of the battery:
    $$R_{Return}\cdot\sum_{Mat}[(1-A_{Mat})\cdot R_{rec,r_Mat}\cdot(E_{recEoL_Mat}-E^{*}_{V_Mat}\cdot\frac{Q_{Sout,r_Mat}}{Q_{P_Mat}})]$$
    - Materials from the non-properly returned fraction of the battery:
    $$(1-R_{Return})\cdot\sum_{Mat}[(1-A_{Mat})\cdot R_{rec,nr_Mat}\cdot(E_{recEoL_Mat}-E^{*}_{V_Mat}\cdot\frac{Q_{Sout,nr_Mat}}{Q_{P_Mat}})]$$
- Impacts and credits of producing secondary materials from the PWB recycling after the battery dismantling, to account for copper, gold, silver and palladium ('electronics recycling'):
$$R_{Return}\cdot[(1-A_{PWB})\cdot E_{recEoL_PWB}-\sum_{Mat}[(1-A_{Mat})\cdot(R_{rec,r_Mat}\cdot E^{*}_{V_Mat}\cdot\frac{Q_{Sout,r_Mat}}{Q_{P_Mat}})]]$$
- Impacts and credits of producing secondary materials from the battery cell recycling, such as copper sulphate, nickel sulphate and cobalt sulphate in the default End of Life and recycling life cycle stage ('cell recycling') of LIB:
$$R_{Return}\cdot\left((1-A_{Batterycell})\cdot E_{recEoL_Batterycell}\right)+R_{Return}\cdot\sum_{Mat}[(1-A_{Mat})\cdot R_{rec,r_Mat}\cdot(E_{recEoL_Mat}-E^{*}_{V_Mat}\cdot\frac{Q_{Sout,r_Mat}}{Q_{P_Mat}})]$$

  Note: This term applies to any material targeted by any cell recycling process, and is therefore not limited to the materials mentioned exemplarily above for the default LIB cell recycling process. For the cell-chemistry-specific default recycling processes, the parameters given in Table 2 apply. If an own company-specific cell recycling process is modelled, it also applies to the corresponding recovered materials not listed in the default recycling process.

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— Impacts due to energy recovery of the plastic from the dismantling of the properly returned battery waste ('energy recovery'):

$$R_{Return} \cdot \sum_{Mat} [(1 - B) \cdot R_{3,r_Mat} \cdot E_{ER_Mat}]$$

— Impacts of disposal ('disposal'):

  - PWB, polymers, battery cell, and other materials not-being recycled, deriving from the dismantling of the non-properly returned battery waste:

$$(1 - R_{Return}) \cdot \sum_{Mat} [(1 - R_{rec,nr_Mat}) \cdot E_{D_Mat}]$$

  - Other materials not-being recycled, deriving from the dismantling of the properly returned battery waste

$$R_{Return} \cdot \sum_{Mat} [(1 - R_{rec,r_Mat} - R_{3,r_Mat}) \cdot E_{D_Mat}]$$

Where:

a) A_{Mat} is the material-specific allocation factor of burdens and credits between two life cycles, the one supplying and the one using recycled material, aiming to reflect market realities. The values for A_{Mat} shall be taken from Table 2. In case a material is not included in Table 2, reference application-specific values shall be taken from Part C of Annex II to Commission Recommendation (EU) 2021/2279 (European Commission, 2021). If no such values for a specific application are there available, then material-specific values for A_{Mat} may be used from the same source. If such values of parameter A_{Mat} for the considered materials are not available either, a value of 0.5 shall be used;

b) $A_{Batterycell}$ is the battery-specific allocation factor and shall be equal to 0.2;

c) A_{PWB} is the PWB-specific allocation factor and shall be equal to 0.2;

d) B is the allocation factor of energy recovery processes and shall be equal to 0;

e) R_{1_Mat} is the material-specific recycled content, meaning the proportion of material in the input to the production that has been recycled from a previous system. R_{1_Mat} is by default equal to 0 for all the materials, unless evidence for a specific value is provided in the carbon footprint study, based on traceability of the supply-chain. Acceptable evidence shall include the documentation referred to in Article 8(1) of Regulation (EU) 2023/1542. Specific values based on supply market statistics are not acceptable;

f) R_{Return} is the battery return rate, meaning the rate of EoL batteries that are properly returned for removal of parts and components multiplied with a 100% collection of contained batteries to be recycled in accordance with Regulation (EU) 2023/1542. R_{Return} is equal to 0.8 by default, unless evidence is provided in the carbon footprint study for a different company-specific value. This may only be done for the share of batteries covered by an ownership business model where the property of the battery stays with the manufacturer and evidence of such ownership model shall be provided in the carbon footprint study together with evidence of the higher return rate, including evidence that the batteries covered under that rate will be recycled in accordance with Regulation (EU) 2023/1542;

Note: The return rate represents the percentage of corresponding batteries that are recovered for recycling in Europe at the end of their life and that are treated with the corresponding recycling processes under EU legislation. The non-returned fraction is of unknown whereabout (e.g., batteries that leave the boundaries of the EU and that do not fall under the corresponding collection schemes).

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Note: The value of 0.8 for the return rate is close to the collection targets for LMT batteries set in Regulation (EU) 2023/1542 (fraction of batteries available for collection), while at the same time aligned with the value for EV batteries, thus keeping consistency.

g) R_{rec,r_Mat} is the material-specific recycling yield for the properly returned fraction of batteries, meaning the proportion of the material in the properly returned fraction of batteries that will be recycled in a subsequent system taking into consideration only the recycling yield. It does not include return rate and dismantling efficiency. Default values for R_{rec,r_Mat} are provided in Table 2. Different company-specific values may be used for R_{rec,r_Mat} in case a company-specific battery cell recycling process is applied and the corresponding evidence is provided in the carbon footprint supporting study. In case such company-specific values are applied, they shall be calculated as the ratio between the mass of the secondary material produced and the mass of the material entering the recycling process. In the case of copper, gold, silver or palladium from the electronics/PWB recycling, R_{rec,r_Mat} refers to the kg of such material per kg of PWB input and only the default values provided in Table 2 shall be used;

h) R_{rec,nr_Mat} is the material-specific recycling yield for the non-properly returned fraction of batteries, meaning the proportion of the material in the non-properly returned fraction of batteries that will be recycled in a subsequent system taking into consideration only the recycling yield. It does not include return rate and dismantling efficiency. The values for R_{rec,nr_Mat} shall be taken from Table 2;

i) R_{3,r_Mat} is the proportion of the material that is used for energy recovery at the EoL of the properly returned waste batteries. For polymers R_{3,r_Mat} shall be 100%. For other materials, R_{3,r_Mat} shall be 0 unless evidence for a different value is provided in the carbon footprint study;

j) E_{V_Mat} is the specific emissions and resources consumed arising from the acquisition and pre-processing of primary material;

k) $E_{recycled_Mat}$ is the specific emissions and resources consumed arising from the recycling process of the recycled material. Collection, sorting, and transportation of the waste used to produce the secondary material are excluded. The allocation of a multi-material process to the single material shall follow the allocation hierarchy set out in Section 6.2. It shall include the management of the waste of the recycling process and the wastewater treatment when applicable. In case $E_{recEoL_Batterycell}$ is modelled with a company-specific process taking place outside Europe, the geographical scope of $E_{recycled_Mat}$ data shall be changed accordingly.

Note: $E_{recycled_Mat}$ can be either a most relevant process or a non-most relevant process, depending on the material. The materials classified as most relevant are provided in Section 5.2.

l) $E_{recEoL_Batterycell}$ is the specific emissions and resources consumed arising from the battery cell recycling. $E_{recEoL_Batterycell}$ is accounted for on battery level and not per material. $E_{recEoL_Batterycell}$ shall be modelled using the default values provided in Table 3 and Table 4 for lithium-ion and similar battery chemistries that are based on closed battery cells with liquid electrolytes operating at room temperature (such as sealed pouch, round cell or prismatic cells). For all other battery chemistries, no specific cell recycling process is available and the battery cells are assumed to be sent to landfill.

Different company-specific recycling processes may be used only for the share of the batteries for which contractual evidence is provided that they will be recycled in a specific recycling plant and that the specific recycling process corresponds to the battery model for which the carbon footprint is declared. In this case, the company-specific battery cell recycling process shall refer to the recycling plant operating at the time of the battery being

placed on the market. The mass and composition of the battery used for the company-specific battery cell recycling process shall reflect the mass and composition of the battery for which the carbon footprint is declared. The justification for modelling a company-specific battery cell recycling process shall be provided in the public version of the carbon footprint study. In case a company-specific battery cell recycling process is applied, the options in Section 5.1 shall apply, to be read as applying to the recycler rather than the supplier. The default recycling process is assumed to take place in Europe, and the geographical region of Europe shall be used for modelling and selection of secondary datasets;

m) E_{recEoL_Mat} is the specific emissions and resources consumed arising from any additional recycling process needed to produce secondary material. In the default battery recycling process, data use shall reflect recycling processes taking place in Europe, both for the properly returned waste batteries and the non-properly returned waste batteries. In case $E_{recEoL_Batterycell}$ is modelled with a company-specific process taking place outside Europe, the geographical scope of E_{recEoL_Mat} data in the cell recycling shall be changed accordingly. For all the outputs of the battery cell recycling process, E_{recEoL_Mat} includes all the additional recycling processes not included in the battery cell recycling, such as sorting and re-melting of metal waste from the casing into secondary metals. It is equal to zero for the default battery cell recycling process since the obtained products in this default process do not require further post-processing;

Note: $E_{recEoL_Batterycell}$ includes all process inputs and direct emissions of the battery cell recycling process. E_{recEoL_Mat} are the specific burdens (carbon footprint) associated with the re-processing of the metal fractions separated during recycling, such as sorting and re-melting of aluminium scrap into secondary aluminium. E_{recEoL_Mat} is zero for all materials that leave the battery cell recycling process as final product i.e., directly replacing the equivalent primary product such as cobalt sulphate or nickel sulphate as in the default EoL life cycle stage. E_{recEoL_Mat} is not zero for any output of the battery cell recycling process that needs additional processes before being sold as secondary material. (e.g., sorting, cleaning and remelting to metal ingots).

n) E_{recEoL_PWB} is the specific emissions and resources consumed arising from the electronics/PWB recycling following the dismantling of the properly returned waste batteries. E_{recEoL_PWB} shall be determined in accordance with Section 5.2.2;

o) $E^*_{V_Mat}$ is the specific emissions and resources consumed arising from the acquisition and pre-processing of primary material assumed to be substituted by recyclable materials. $E^*_{V_Mat}$ shall be modelled based on the European average production or, when the European average production is not available, based on the global average production. However, $E^*_{V_Mat}$ shall be equal to E_{V_Mat} when E_{V_Mat} is lower than the applied average production;

p) Q_{P_Mat} is the quality of the primary material;

q) Q_{Sin_Mat} is the quality of the ingoing secondary material, meaning the quality of the recycled material at the point of substitution. Values for Q_{sin} / Q_p shall be taken from Table 2;

r) $Q_{Sout,r}$ and $Q_{Sout,nr}$ are the quality of the outgoing secondary material, meaning the quality of the recyclable material at the point of substitution, for the properly returned and the non-properly returned fraction, respectively. Default values for $Q_{Sout,r} / Q_p$ and $Q_{Sout,nr} / Q_p$ are provided in Table 2. Higher values are allowed only if a company-specific battery cell recycling process is modelled and if evidence is provided in the carbon footprint study, such as through technical specifications of the secondary material demonstrating that is 'battery-grade';

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s) E_{ER_Mat} is the specific emissions and resources consumed arising from the energy recovery of the polymers from the battery waste dismantling, for both the properly returned and the non-properly returned waste batteries, including both the direct emissions of the incineration plants and the credits due to the avoided energy production. E_{ER_Mat} shall be determined in accordance with Section 5.2.2. If the secondary datasets for energy recovery do not include the credits from the production of energy, the credits shall be modelled using the lower heating value of the polymers and by substituting the European energy mix;

Note: For this purpose, if incineration with energy recovery is modelled for such substances, check if the secondary dataset used for modelling the incineration process already includes the credits for recovered energy. If it does not, then the credit shall be modelled based on the lower heating value by crediting the corresponding amount of energy, using the dataset for the corresponding average European energy mix.

t) E_{D_Mat} is the specific emissions and resources consumed arising from the disposal of waste material at the analysed product's EoL, without energy recovery.

- If the datasets used for PWB recycling include the credits from the production of secondary material, the credits for specific materials shall be set to zero.
- If the used datasets already include the allocation with the factor A_{Mat}, then they shall not be allocated again.
- To apply the CFF correctly along the battery supply-chain, the company-specific R_1 of e.g., any raw material, precursors, active material, shall be handed over to the next process step / manufacturer even in case of CFB-compliant company-specific datasets.

Note: When using an R_1 (recycled content) different from zero i.e., if secondary material is used as input for a process in the manufacturing life cycle stage, and if the secondary material is a product of a most relevant process as defined in Section 5.2, it may be modelled using company-specific data according to Section 5.2.1. In this case, the waste product (e.g., waste battery) that is processed by the recycling process for obtaining the secondary materials is burden free according to the EF Recommendation (2279/2021). The provider of the corresponding secondary material may then provide a company-specific dataset according to the present rules for the secondary material(s) obtained from its process. These datasets will be used as gate-to-gate processes that produce secondary materials ($E_{recycled_Mat}$).

Note: If an own, company-specific recycling process is modelled where components are recycled on a higher level i.e., not broken down to individual materials but recovered as complex products, the CFF shall be applied per material. Complex product refers to components that are not recovered as individual materials (or elements), but on a higher level of complexity, such as in the case of re-use of components like battery trays, or for direct recycling of cathode active materials (CAM) where the CAM is not decomposed for recovering its constituents (cobalt sulphate, nickel sulphate, etc.), but recovered integrally for re-use in the production of new battery cathodes. In this case, on the output side, credit may be given by material for $E^*_{V_Mat} = E_{V_Mat} + E_{gtg\text{-}c}$ i.e., for the avoided primary material plus the corresponding gate-to-gate impacts of component manufacturing ($E_{gtg\text{-}c}$). For the material-specific application of the CFF on the input side, a dataset shall be provided by the entity that produces the recycled complex product (such as the recycled CAM) for each of the materials that compose the complex product, giving the material-specific $E_{recycled_Mat}$. Allocation of the recycling process burdens (E_{recEoL_Mat}) to each of the materials that constitute the recycled complex product shall be done according to Section 6.2.

Table 2. Default values for the parameters for the EoL and recycling life cycle stage

				For the properly returned waste batteries		For the non-properly returned waste batteries	
	A_{Mat}	R_{1_Mat}	Q_{Sin}/Q_p	R_{rec,r_Mat}	$Q_{Sout,r}/Q_p$	R_{rec,nr_Mat}	$Q_{Sout,nr}/Q_p$
Al metal (from dismantling)	0.2	0 (a)	1	0.9 (b)	1 (b)	0.9	1
Al metal (cells)	0.2	0 (a)	1	0 (b)	1 (b)	0	1
Cu metal (from dismantling)	0.2	0 (a)	1	0.9 (b)	1 (b)	0.9	1
Cu metal (cells)	0.2	0 (a)	1	0.9 (b)	1 (b)	0	1
Fe metal (from dismantling)	0.2	0 (a)	1	0.9 (b)	1 (b)	0.9	1
Fe metal (cells)	0.2	0 (a)	1	0 (b)	1 (b)	0	1
Polymers (from dismantling)	0.5	0 (a)	1	0 (b)	0.8 (b)	0	0.8
Other materials (from dismantling)	0.5	0 (a)	1	0 (b)	n/a (b)	0	n/a
Au from PWB	0.2	0 (a)	1	$1.4x10^{-5}$ (c)	1	0	1
Cu from PWB	0.2	0 (a)	1	0.11 (c)	1	0	1
Ag from PWB	0.2	0 (a)	1	$9.77x10^{-4}$ (c)	1	0	1
Pd from PWB	0.2	0 (a)	1	$9.31x10^{-8}$ (c)	1	0	1
Co compounds (cell)	0.2	0 (a)	1	0.9 (b)	0.8 (b)	0	0.8
Ni compounds (cell)	0.2	0 (a)	1	0.9 (b)	0.8 (b)	0	n/a
Mn compounds (cell)	0.2	0 (a)	1	0 (b)	0.8 (b)	0	n/a
Li compounds (cell)	0.2	0 (a)	1	0 (b)	0.8 (b)	0	n/a
Na compounds (cell)	0.2	0 (a)	1	0 (b)	0.8 (b)	0	n/a
Tin	0.2	0 (a)	1	0 (b)	1 (b)	0	n/a
Other metals and metal compounds	0.2	0 (a)	1	0 (b)	0.8 (b)	0	n/a
Graphite / hard carbon (cell)	0.2	0 (a)	1	0 (b)	0.8 (b)	0	n/a
Other materials (cell)	0.5 (d)	0 (a)	1 (d)	0 (b)	0.8 (b)	0	n/a

(a) Company-specific values may be used under the conditions set out in this section.
(b) Company-specific values shall be used if a company-specific battery cell recycling is used.
(c) Value shall be applied per kg of recycled electronics/PWB.
(d) Additional materials may be accounted for if a company-specific battery cell recycling process is modelled.
n/a = not applicable.

Source: JRC own elaboration

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Table 3. Default cell recycling model, inventory data per kg of battery cell, pyrometallurgical treatment. Equal to the CFB-EV

Component	Unit	Default value	Specification / dataset
Inputs			
EoL battery cells	kg	1.00	From disassembly
Electricity	kWh	1.24	Electricity mix
Process heat	MJ	0.360	Thermal energy from natural gas
Process heat	MJ	0.272	Thermal energy from diesel fuel
Limestone washed	kg	0.150	Limestone production, technology mix
Silica sand	kg	0.186	Silica sand, production mix
Quicklime (CAO)	kg	0.111	Quicklime production, technology mix
Wet lime ($Ca(OH)_2$)	kg	0.010	Calcium hydroxide production; technology mix
Carbon black	kg	0.001	Carbon black production, technology mix
Water	kg	0.298	Tap water, average technology mix
Transport from the disassembling to the battery cell recycling plant	km	130	Transport by truck (>32 t, EURO 4)
Transport from the disassembling to the battery cell recycling plant	km	240	Transport by train (average freight train)
Transport from the disassembling to the battery cell recycling plant	km	270	Transport by ship (barge)
Outputs			
Metal alloy	kg	0.209	To hydrometallurgical treatment
Slag	kg	0.952	Landfill of inert slag
Emissions	kg CO_2-eq.	1.051	Direct process emissions (all greenhouse gases)

Source: JRC analysis, based on PEFCR on rechargeable Batteries (RECHARGE, 2024).

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Table 4. Default recycling model, inventory data per kg of battery cell, hydrometallurgical treatment

Component	Unit	Default value	Specification / dataset
Inputs			
Metal alloy	kg	0.209	
Electricity	kWh	2.59	Electricity mix
Process heat	MJ	4.79	Thermal energy from natural gas
Hydrochloric acid (100%)	kg	0.008	HCl production, technology mix
Hydrogen peroxide (100%)	kg	0.274	Hydrogen peroxide, production, technology mix
Soda (sodium carbonate)	kg	0.017	Soda production, technology mix
Sodium hydroxide (100%; caustic soda)	kg	0.42	Sodium hydroxide production, technology mix
Sulphuric acid aq. (96%)	kg	0.84	Sulphuric acid production, technology mix
Water (tap water)	m^3	0.01	
Outputs			
Recovered metals such as aluminium, copper, steel and stainless steel	kg		Credit for each specific metal according to its content in the battery cell and parameters R_{Return}, A_{Mat}, R_{rec,r_Mat}, and $Q_{Sout,r}/Q_p$
Recovered metal salts such as cobalt sulphate and nickel sulphate	kg		Credit for each specific metal salt according to its stoichiometric content in the battery cell and parameters R_{Return}, A_{Mat}, R_{rec,r_Mat} and $Q_{Sout,r}/Q_p$
Wastewater	m^3	$8.64x10^{-3}$	Wastewater treatment, average

Source: JRC analysis, based on PEFCR on rechargeable Batteries (RECHARGE, 2024)

6.3.2 The circular footprint formula ('CFF') applied to manufacturing waste

Identical to CFB-IND.

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6.4 Transport

Identical to CFB-IND.

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7 Verification

7.1 Documentation

7.1.1 Public version of the CFB supporting study

The public version of the CFB study shall give meaningful information to consumers and other end-users on the calculation of the carbon footprint and shall include:

- information about the battery model;
- information about the geographical location of the battery manufacturing plant;
- the total life cycle carbon footprint expressed in kg CO_2-equivalent per kWh of total energy delivered;
- the reference year of the carbon footprint calculation i.e., the year for which the data for the CFB model was collected;
- the DQR score and the values of the single DQR criteria calculated in accordance with Section 5.5;
- the initial useable energy capacity, and the final useable energy capacity of the battery in kWh and its power in kW;
- the total amount of CO_2-equivalent emitted during the life cycle of the battery, expressed in kg CO_2-equivalent per (unit of) battery, and the total amount of energy delivered by the battery over its service life as used for the carbon footprint calculation, in kWh;
- the application the battery is designed for (if applicable), the corresponding service life in terms of cycles and years (years of operation), including (if applicable) the corresponding justification, and the threshold for reaching the end of service life in percent of the battery's initial useable energy capacity;
- the exact test procedure (including charge-discharge profiles over time, C-rates, temperature), the standard operating conditions and, if applicable, the underlying standard used for determining the battery lifetime;
- information on each dataset used: the name of the dataset, for which life cycle stage or stages and which process or processes it was used, whether it is a company-specific dataset or a secondary dataset, its source in case of secondary datasets, its values of the DQR criteria, and its time validity;
- information on electricity modelling:
    - since for electricity modelling this report is aligned with the CFB-IND, which in turn refers to the methodology specified in the final Delegated Act establishing the methodology for the calculation and verification of the carbon footprint of electric vehicle batteries, it is recommended to align with it also regarding the requirements for description and documentation of the electricity modelling;
- information on allocation:
    - indication of the multifunctional processes for which an allocation was applied;
    - which point of the hierarchy in Section 6.2.1 was applied in each case and a justification why any earlier points in the hierarchy could not be applied;
- information on recycled content and EoL modelling:

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- in case of recycled content higher than zero, information on the recycled content share;
- in case of a company-specific return rate, information on the share of the batteries that are covered by an ownership business model;
- the justification for modelling a company-specific battery cell recycling process, where applied.

Note: This section will still be refined according to the final choice on the functional unit.

Note: useable energy capacity is the energy available to the user when discharging a fully charged, new battery (at beginning of its service life) within the charge/ discharge limit set by the battery management system under standard operating conditions i.e., taking into account both charge limits and discharge limits.

7.1.2 Non-public version of the CFB supporting study

The carbon footprint study shall document in a systematic, orderly and comprehensive manner all steps taken for calculating the carbon footprint.

It shall include detailed description and documentation of:

- the carbon footprint model used for the calculation;
- any cut-off applied in the modelling, the resulting mass balance gap and an indication to which activity data or dataset the missing mass is assigned;
- the functional unit and all the information needed to calculate it;
- the details of all the company-specific data, including:
  - one or more tables containing all LCI, activity data and elementary flows, the corresponding background dataset used, if applicable, and the values of the DQR criteria of each dataset ('inventory tables'), both per functional unit and per kg of product. The tables shall include the list of parameters provided in Section 5.1, including the bill of materials or ingredients, such as substance names, units and quantities, information on origin, grades or purities, yield rates, where applicable, recycled content and other technically or environmentally relevant characterisation, the procedure used for company-specific data collection, estimation and calculation, the data source where applicable, the data collection period, and a justification if data collection is done for periods of time other than the default timeframes specified in this document;
  - to which life cycle stage the data belong;
  - an indication of whether the data have been used for modelling the mandatory company-specific or non-mandatory company-specific processes;
  - all data sources, mathematical treatments applied to the data, and any underlying documentation needed to establish the reliability of the company-specific data.
- the details of all secondary and company-specific datasets used, including:
  - for non-mandatory most relevant processes, whether secondary or company-specific datasets were used, the method of selecting the datasets including any relevant assumptions and justifications such as the choice of a proxy in the case of TeR equalling four, which datasets were used, whether the electricity in the '-1 level'

was changed and, in case it was changed, the justification for such change. In case company-specific datasets were used for non-mandatory most relevant processes in the raw material acquisition and pre-processing life cycle stage, the reference flow, the specific concentration of the target element (if applicable), and other substances contained as impurities in the main product (target material), in concentration or specific content of relevant elements, to allow a proper mass balance check;

- for all secondary datasets the exact name of the dataset, the source, its universal unique identifier (UUID), location, dataset type ('LCI result' or 'partially terminated'), the values of the DQR criteria, and the time validity;
- for all the company-specific datasets, the exact name of the dataset, its UUID, location, dataset type ('LCI result' or 'partially terminated'), the values of the DQR criteria, the time validity and the corresponding documentation for verification;

— the details of how the carbon footprint of electricity use was determined:

- since for electricity modelling this report is aligned with the CFB-IND, which in turn refers to the methodology specified in the final Delegated Act establishing the methodology for the calculation and verification of the carbon footprint of electric vehicle batteries (still under finalisation at the date of the publication of the present draft report), it is recommended to align with it also regarding the requirements for description and documentation of the electricity modelling.

— the details of how allocation was applied, including:

- indication of the multifunctional processes for which an allocation was applied and the allocation factors used;
- which point of the hierarchy in Section 6.2.1 was applied in each case with a justification of why any earlier points in the hierarchy could not be applied;
- in case of allocation based on physical relationships, which physical relationship has been used, a justification thereof, and the underlying data sources;
- in case of economic allocation, the prices and the corresponding sources including the considered timespans used for the economic allocation, and justification for any timespans applied that are shorter than the default ones;
- the datasets or activity data to which the allocation factors apply shall be indicated in the inventory tables;
- if allocation is done for energy and auxiliary inputs of production lines as referred to in Section 6.2.2, the method applied and a justification thereof;

— the details of the recycled content and EoL modelling, including:

- the parameters used in the CFF for all materials, with evidence for the cases where values other than the default values were used;
- in case of recycled content higher than zero, the evidence for the recycled content share;
- in case of a company-specific return rate, the evidence for the share of the batteries that are covered by an ownership business model where the property of the battery stays with the manufacturer together with evidence of the higher return rate, including evidence that the batteries covered under that rate will be recycled in accordance with Regulation (EU) 2023/1542;

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    - in case of a company-specific battery cell recycling process, the evidence that the batteries will be recycled in a specific recycling plant and that the recycling process corresponds to the battery model for which the carbon footprint is declared;
- the categories of the transport means and the estimated distances;
- the public version of the carbon footprint study.

Note: This section will still be refined according to the final choice on the functional unit.

7.2 Verification and validation techniques

Identical to CFB-IND.

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8 Conclusions

The present report constitutes an initial proposal for the framework for determining the carbon footprint of rechargeable LMT batteries. It is based on the JRC report that provides the methodological guidelines on the carbon footprint for industrial batteries without external storage (CFB-IND) and follows a delta approach, with only those Sections developed in detail where relevant discrepancies are given.

As such, it is conceived as a technical basis for exchange and stakeholder feedback towards a final report on the CFB Rules for LMT Batteries and for the corresponding Delegated Act as under the Article 7 of the Batteries Regulation. JRC explicitly welcomes any feedback on the suggested methodology and will consider all input for updating this draft towards a final Science for Policy Report.

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List of abbreviations

AAM	Anode Active Material
CAM	Cathode Active Material
CFB	Carbon Footprint of Batteries
CFB-EV	Carbon Footprint of Electric Vehicle Batteries
CFB-IND	Carbon Footprint of Industrial Batteries without external storage
CFF	Circular Footprint Formula
DA-EV	Delegated Act on carbon footprint of EV batteries
DQR	Data Quality Rating
EF	Environmental Footprint
EoL	End of Life
EV	Electric Vehicle
FEqC	Full Equivalent Cycles
ILCD	International Reference Life Cycle Data system
JRC	Joint Research Centre
LCA	Life Cycle Assessment
LCDN	Life Cycle Data Network
LCI	Life Cycle Inventory
LIB	Lithium-Ion Battery
LMT	Light Means of Transport
PEF	Product Environmental Footprint
PEFCR	Product Environmental Footprint Category Rules
PWB	(populated) Printed Wiring Board

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List of definitions

'Active material' refers to the materials within the battery cell which react chemically to produce electric energy when the battery cell discharges or to store electric energy when the battery is being charged. They are the main constituents of the anode and cathode, respectively. For lithium-ion batteries, anode active materials are typically graphites, sometimes doped with silicon, or spinels such as lithium titanate. Cathode active materials (CAM) are typically layered oxides such as Lithium Nickel Manganese Cobalt Oxide or Lithium Nickel Cobalt Aluminium Oxide, but also polyanionic materials such as Lithium Iron Phosphate.

'Active material precursor' means material required for the synthesis or production of active materials, either cathode active material precursors or anode active material precursors.

'Activity data' means the information associated with processes while modelling Life Cycle Inventories (LCI). The aggregated LCI results of the process chains that represent the activities of a process are each multiplied by the corresponding activity data and then combined to derive the carbon footprint associated with that process (Annex II Regulation (EU) 2023/1542).

'Battery' means any device delivering electrical energy generated by direct conversion of chemical energy, having internal or external storage, and consisting of one or more non-rechargeable or rechargeable battery cells, modules or of packs of them (Article 3 of Regulation (EU) 2023/1542).

'Battery cell' means the basic functional unit in a battery constituted by electrodes, electrolyte, container, terminals and, if applicable, separators, and containing the active materials the reaction of which generates electrical energy (Article 3 of Regulation (EU) 2023/1542).

'Battery management system' means an electronic device that controls or manages the electric and thermal functions of a battery in order to ensure the battery's safety, performance and service life, manages and stores the data for the parameters for determining the battery's state of health and expected lifetime set out in Annex VII and communicates with the vehicle, light means of transport or appliance in which the battery is incorporated, or with a public or private charging infrastructure (Article 3 of Regulation (EU) 2023/1542).

'Battery manufacturing waste' means the materials or objects rejected during the battery manufacturing process, which cannot be re-used as an integral part in the same process and need to be recycled (Article 3 of Regulation (EU) 2023/1542).

'Battery pack' means any set of battery cells or modules that are connected together or encapsulated within an outer casing, to form a complete unit which is not meant to be split up or opened by the end-user (Article 3 of Regulation (EU) 2023/1542).

'Battery model' means a version of a battery all units of which share the same technical characteristics relevant for the requirements of this Regulation on sustainability, safety, labelling, marking and information, and the same model identifier (Article 3 of Regulation (EU) 2023/1542).

'Beginning of service life' means the moment when a new battery starts its operation i.e., just before it provides its first charge-discharge cycle within the application.

'Bill of materials' means a list of the raw materials, sub-assemblies, intermediate assemblies, sub-components and parts, and the quantities of each, needed to manufacture the battery (Regulation (EU) 2023/1542 - Annex II).

'Carbon footprint' means the sum of greenhouse gas emissions and greenhouse gas removals in a product system, expressed as carbon dioxide equivalents and based on a Product Environmental Footprint (PEF) study using the single impact category of climate change (Article 3 of Regulation (EU) 2023/1542).

'CFB-EV' indicates, in this report, the value of the carbon footprint to be declared for electric vehicle batteries, according to Article 7 of the Regulation (EU) 2023/1542.

'CFB-IND' indicates, in this report, the methodological guidelines needed to calculate the value of the carbon footprint of industrial batteries without external storage, as published in the specific JRC Science for Policy Report since April 2025.

'CFB-compliant company-specific dataset' means a company-specific dataset that respects the requirements defined in Section 5.3 of this document. A CFB-compliant company-specific dataset can be either a LCI dataset or a LCIA dataset.

'Characterisation factor' means a factor derived from a characterisation model which is applied to convert an assigned life cycle inventory result to the common unit of a LCIA impact category.

'Circular Footprint Formula' describes how burdens and benefits from disposal and recovery of the product or service assessed, as well as use of secondary materials (i.e., recycled content) into that product or service, are allocated to the system under study.

'Climate change' is the LCIA impact category considering all inputs and outputs that result in greenhouse gas (GHG) emissions. The consequences include increased average global temperatures and sudden regional climatic changes.

'Company-specific dataset' means a dataset (disaggregated or aggregated) compiled with company-specific data. The activity data and direct elementary exchanges of company-specific data sets shall refer to company-specific information, while the underlying sub-processes may be modelled with company-specific or secondary datasets.

'Dataset' means a document or file with life cycle information of a specified product or other reference such as a process, covering its descriptive metadata and either its quantitative life cycle inventory in case of a life cycle inventory ('LCI') result dataset or its carbon footprint in case of a life cycle impact assessment ('LCIA') result dataset.

'Data collection period' means the date(s) or time period(s) when the data was collected. Note that this does NOT refer to e.g., the publication dates of papers or books from which the data may stem, but to the original data collection period.

'EF-compliant dataset' means dataset developed in compliance with the EF requirements in terms of modelling and methodological compliance (in agreement with the "Commission Recommendation (EU) 2021/2279"); meta data compliance (in agreement with Fazio et al. 2020, "Guide on EF compliant datasets"); nomenclature, and characterization factors in agreement with the "EF 3.1 Reference Package"). The DQR of each single data quality indicator shall be lower or equal than 3.

'Elementary flows' means the materials or energy entering the system being studied that have been drawn from the environment without previous human transformation, or material or energy leaving the system being studied that is released into the environment without subsequent human transformation.

'End of service life' means the state of the battery where it is, due to ageing or other degradation processes, considered unsuitable for continuing to provide service within the primary application and where it is decommissioned or discarded and either sent for recycling or for other treatment (such as refurbishing, remanufacturing, preparation for re-use or preparation for repurposing) for subsequent use in a second life application.

'Final useable energy capacity' means the energy available to the user when discharging a fully charged battery at its end of service life within the charge/ discharge limit set by the battery management system under standard operating conditions (such as maximum depth of discharge for increasing lifetime). Fully charged refers to charging until the charge level set by the battery management system under standard operating conditions.

'Full cycles' is the number of times the battery can fully discharge within the charge/discharge limits set by the battery management system under standard operating conditions. The actual energy delivered in each full cycle decreases over lifetime. For a new battery (at beginning of its service life), one full cycle is equivalent to one full equivalent cycle (FEqC). The full cycles multiplied with the initial useable energy capacity and the average value between 100% and the state of health (or remaining useable energy capacity in %) at EoL gives the total energy provided by the battery over lifetime.

'Full equivalent cycles' (FEqC) is the number of times the battery would theoretically be able to discharge its initial useable energy capacity. The FEqC multiplied with the initial useable energy capacity gives the total energy provided by the battery over lifetime.

'Functional unit' means the qualitative and quantitative aspects of the functions, services, or both, provided by the battery (Annex II of Regulation (EU) 2023/1542).

'ILCD entry-level compliant dataset' means a dataset developed in accordance with the compliance rules and entry-level requirements of the International Reference Lifecycle Data System ('ILCD') Data Network, available on the European Platform on LCA, and in accordance with the EF Reference Package, available in the Life Cycle Data Network ('LCDN') on the European Platform of LCA, for nomenclature and characterisation factors compliance.

'Initial useable energy capacity' means the energy available to the user when discharging a fully charged, new battery (i.e., at the beginning of its service life) within the charge/ discharge limits set by the battery management system under standard operating conditions (such as maximum depth of discharge for increasing lifetime). "Fully charged" refers to charging until the charge level set by the battery management system under standard operating conditions.

'Intermediate product' means the output form of a unit process that in turn is input to other unit processes which require further transformation within the system. An intermediate product is a product that requires further processing before it is saleable to the final consumer.

'LCI result dataset' means a dataset modelling a complete or partial life cycle of a product system that, apart from the product flow associated with the reference product, lists only elementary flows.

'LCIA result dataset' means a dataset modelling a complete or partial life cycle of a product system that, apart from the product flow associated with the reference product, lists only the result of the life cycle impact assessment, which in the case where climate change is the only impact category analysed means that the result is the amount of carbon dioxide ('CO_2')-equivalent.

'Life cycle' means the consecutive and interlinked stages of a product system, from raw material acquisition or generation from natural resources to final disposal (ISO 14040:2006 or equivalent) (Annex II of Regulation (EU) 2023/1542).

'Life cycle inventory (LCI)' means the combined set of exchanges of elementary, waste and product flows in a LCI dataset (Annex II of Regulation (EU) 2023/1542).

'Life cycle inventory (LCI) dataset' means a document or file with life cycle information on a specified product or other reference, such as the site or process, covering descriptive metadata and quantitative life cycle inventory, which could include a unit process dataset, partially disaggregated or an aggregated dataset.

'Lifetime of a battery' means the period that starts when the battery is manufactured and ends when the battery becomes waste (Article 3 of Regulation (EU) 2023/1542).

'Manufacturer' means any natural or legal person who manufactures a battery or has a battery designed or manufactured, and markets that battery under its own name or trademark or puts it into service for its own purposes (Article 3 of Regulation (EU) 2023/1542).

'Most representative secondary dataset' means the secondary dataset having the highest technological representativeness ('TeR') quality rating or, if there are several datasets with the same TeR, the one with the highest geographical representativeness ('GeR') quality rating, or, in the absence of information on the geographical provenience of the material concerned, the one representing the global average.

'Notified body' means a conformity assessment body that has been notified in accordance with Chapter V of Regulation (EU) 2023/1542.

'Partial cycles' is the number of times the battery can discharge with a given discharge profile, but without fully discharging within the charge/discharge limits set by the battery management system.

'Partially disaggregated dataset' means a dataset with a LCI that contains elementary flows and activity data, and that only in combination with its complementing underlying datasets yields a complete aggregated LCI dataset.

'Partially disaggregated dataset at -1 level' means a partially disaggregated dataset that contains elementary flows and activity data for one level down in the supply-chain, while all complementing underlying datasets are in their aggregated form.

'Placing on the market' means the first making available of a battery on the Union market (Article 3 of Regulation (EU) 2023/1542).

'Recycler' means any natural or legal person who carries out recycling in a permitted facility (Article 3 of Regulation (EU) 2023/1542).

'Reference flow' means the measure of the outputs from processes in a given product system required to fulfil the function expressed by the functional unit.

'Secondary data' means data not from a specific process within the supply-chain of the company performing a carbon footprint study. This refers to data that is not directly collected, measured, or estimated by the company, but sourced from a third party LCI database or other sources. Secondary data includes industry average data (e.g., from published production data, government statistics, and industry associations), literature studies, engineering studies and patents, and may also be based on financial data, and contain proxy data, and other generic data.

'Secondary dataset' refers to any dataset that is not company-specific, e.g., from a database.

'State of health' means a measure of the general condition of a rechargeable battery and its ability to deliver the specified performance compared with its initial condition (Article 3 of Regulation (EU) 2023/1542).

'System boundary' indicates which aspects are included or excluded from the life cycle study. The system boundary defines which parts of the product life cycle and which associated life cycle stages and processes belong to the analysed system (i.e., are required for carrying out its function as defined by the functional unit), except for those processes excluded based on the cut-off rule.

'Technical specification' means a document that prescribes technical requirements to be fulfilled by a product, process or service (Article 3 of Regulation (EU) 2023/1542) and that is delivered together with the battery to the customer or end-user.

'Treatment' means any operation carried out on waste batteries after they have been handed over to a facility for sorting, preparation for re-use, preparation for repurposing, preparation for recycling or for recycling (Article 3 of Regulation (EU) 2023/1542).

'Unit process' means the smallest element considered in the LCI for which input and output data are quantified (based on ISO 14040:2006).

Apart from the previously listed, the definitions provided in Regulation (EU) 2023/1542 apply.

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